

Brief Introduction to MRI Physics

1

Joao Tourais^a, Chiara Coletti^a, and Sebastian Weingärtner

Department of Imaging Physics, Delft University of Technology, Delft, The Netherlands

1.1 A brief history of MRI

In the 1970s, the State University of New York decided against supporting a patent application, with the rationale that the resulting patent would be unlikely to recoup the patent application costs. The patent application in question was written by Paul C. Lauterbur, describing his technique for creating an image from nuclear magnetic resonance (NMR) using field gradients. Some 30 years later, this invention would be recognized by the Nobel Committee as fundamental to the development of what is now known as magnetic resonance imaging (MRI)—and by the State University of New York, Lauterbur's employer at the time, as a huge missed opportunity. Ever since Lauterbur created his early predecessor to MR images by rotating a probe with deuterium and water in an NMR spectrometer [11], MRI has come a long way. With nearly 100 million scans performed each year worldwide, MRI is one of the backbones of modern image-based diagnosis. A plethora of key innovations by brilliant minds has facilitated the techniques that now enable unique insights into the human body.

MRI is based on the phenomenon of NMR, which was first described by Isidor I. Rabi in the 1930s [17] and later confirmed by experimental validation in 1938 [18]. The next step was taken by Felix Bloch and Edward M. Purcell, who independently demonstrated NMR in liquids and solids in 1946 [2,16]. These discoveries marked the beginning of NMR spectroscopy, which in the following years proved to be a transformative technique for the study of molecules. However, further leaps were required to enable the formation of images.

The first step in this direction was taken by Hermann Y. Carr, who reported in his 1952 Ph.D. thesis the use of spatially varying magnetic fields, so-called gradient fields, to encode the spatial location of magnetization in NMR experiments [4]. In his landmark paper from 1973, Lauterbur built on this idea to create a two-dimensional image, introducing the fundamental concept of image formation used in modern MRI machines [11]. Following Lauterbur's seminal work, Sir Peter Mansfield in Nottingham, UK, introduced the mathematical characterization of the newly formed imaging methodology [13]. The contributions of Paul C. Lauterbur and Sir Peter Mansfield that enabled the development of MRI were honored with the Nobel Prize for Physiology and Medicine in 2003. These developments sparked the rapidly evolving field of MRI research, which remains an active area of research to date.

In this chapter, we describe the basic physical principles that underpin MRI and that are necessary to understand the properties of MR images, data acquisition techniques, and reconstruction methods.

^a Those authors contributed equally to the work.

We take an explorative approach to derive six principles that aim to enable the reader to follow in-depth discussions of MRI: 1) the formation of a magnetization based on nuclear magnetism; 2) the dynamics of this magnetization in an external magnetic field; 3) the use of rotating magnetic fields to create a detectable magnetization component; 4) the use of induction to measure this component; 5) the evolution of this signal in the presence of relaxation mechanisms, and finally 6) the use of gradient fields to enable image formation. By the end of this chapter, the reader will hopefully have attained a basic understanding of an MRI experiment and the various components involved. This forms the foundation necessary to understand and discuss advanced MRI reconstructions.

1.2 Nuclear magnetism

The signal in MRI originates from magnetization that forms within the sample or the subject that is being imaged. The source of this magnetization can be found in the nuclei of certain atoms within the sample, due to the physical phenomenon known as nuclear magnetism. To understand nuclear magnetism and the resulting MRI signal, the basic properties giving rise to this phenomenon and the physical principles describing its behavior are laid out in this section.

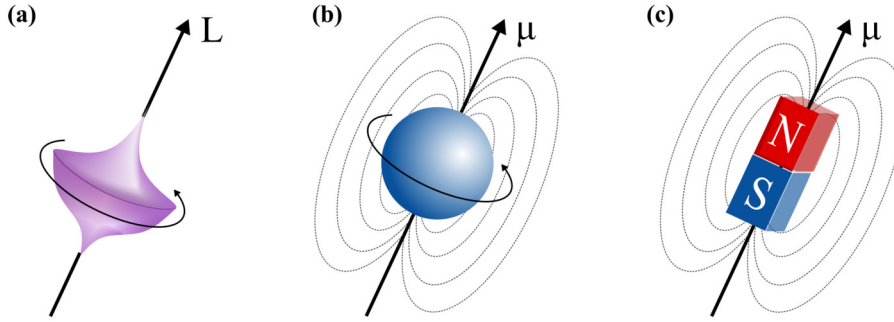
1.2.1 Spin

Particles and nuclei are described by several characteristics, for example, their mass or their charge. Similar to mass or charge, *spin* also describes a fundamental property intrinsic to particles and nuclei. As hinted at by its terminology, a spin is related to an angular momentum, similar to the rotational momentum of a spinning top. However, unlike the classical angular momentum of the spinning top, the spin angular momentum is quantized, i.e., it can only assume certain discrete states.

Think of this as the difference between a bicycle wheel and a fortune wheel: classical angular momentum can align in any direction, similar to a bicycle wheel that can stop at any angle. A spin angular momentum, however, has only a discrete set of directions, similar to the stopping position of the fortune wheel. The number of possible directions, also known as spin states, is characterized by the *spin quantum number*, s . By conventional definition, s is a half or full integer ($s = 0, 1/2, 1, 3/2, \dots$). Using the spin quantum number, the number of different spin states is given by $2s + 1$. The spin quantum number is a constant property that does not change over time and cannot be altered without modifying the particle or the nucleus. The spin state, on the other hand, can change over time and may be different for otherwise identical particles or nuclei.

Going back to our analogy of the spinning top, if additionally an electric charge is placed on the edge of the spinning top, it rotates due to the rotational momentum. Classical electrodynamics tells us that the rotation of the charge gives rise to a magnetic moment oriented along the rotational axis. Similarly, nuclei with nonzero spin develop a magnetic moment. This nuclear magnetic moment is primarily a magnetic dipole moment, which means its magnetic field is comparable to the one created by an infinitely small bar magnet. Thus, a useful analogy of nuclei for the understanding of MRI is a spinning bar magnet as illustrated in Fig. 1.1.¹

¹ The analogy has naturally certain limitations. For example, nuclei with asymmetric charge distribution also exhibit the weak quadrupole moments associated with different resulting magnetic fields. These effects can have relevant consequences on the signal when imaged with MRI. For more details, the reader is referred to [28,21].

**FIGURE 1.1**

(a) Spinning top with angular momentum L , (b) hydrogen nucleus (proton) with magnetic moment μ , and (c) spinning bar magnet with magnetic dipole moment μ .

1.2.2 Net magnetization

When a nucleus that exhibits a nuclear magnetic moment is placed into an external magnetic field $\mathbf{B}$, different orientations of the magnetic moment are associated with different magnetic potential energies. For the discrete spin states, this entails an energy gap between the various states. This splitting of the energy levels is called the *Zeeman effect*. The gap between the energy states depends on the strength of the external magnetic field B , the magnetic moment (μ), and the spin (s). A useful quantity to describe the behavior of nuclei in external magnetic fields is the gyromagnetic ratio that is obtained as the ratio between the magnetic and the angular momentum $\gamma = \mu/(s\hbar)$, where $\hbar$ is the reduced Planck constant. The energy gap between the spin states can then be expressed as

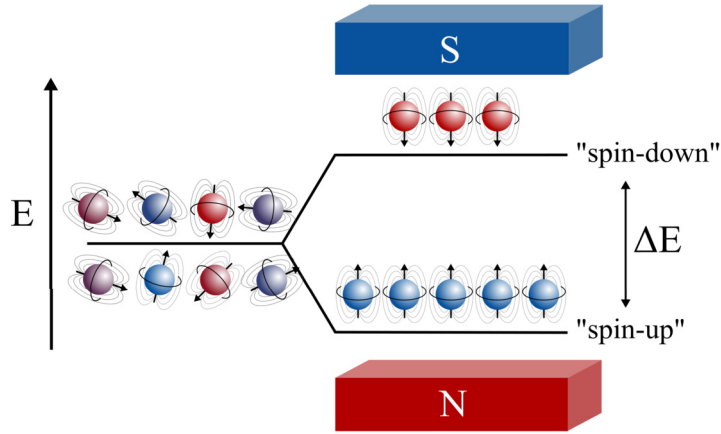
$$\Delta E = \frac{\mu}{s} B = \gamma \hbar B. \quad (1.1)$$

When placing a multitude of nuclei into the external magnetic field, they will naturally distribute across the spin states following the principles of thermodynamics. This distribution can be described using Boltzmann statistics as a function of the energy gap ΔE and the temperature T as

$$\frac{N^+}{N^-} = e^{-\frac{\Delta E}{kT}}, \quad (1.2)$$

with N^+ and N^- being the number of nuclei in the higher and lower energy spin state, respectively. Here, k denotes the Boltzmann constant. The uneven distribution of the orientations of the magnetic moments entails that, when summing all magnetic moments in a volume, a nonzero magnetization is formed.

MRI most commonly focuses on the hydrogen nucleus ^1H which comprises only a single proton. A proton is a fermion, which means it has a half-integer spin, in this case, $s = 1/2$. Accordingly, the hydrogen nucleus has only $2s + 1 = 2$ spin states, which are often referred to as *spin-up* and *spin-down*, respectively (Fig. 1.2). In this case, the net magnetization is easily derived by subtracting the

**FIGURE 1.2**

Zeeman splitting effect: When no external magnetic field is present, spins with randomly distributed magnetic moments are characterized by the same energy level. In the presence of an external magnetic field, the spin-up and spin-down states are separated by a potential energy gap.

anti-parallel (spin-down) magnetic moments from the parallel magnetic moment (spin-up).

$$M_0 = N^+ \mu - N^- \mu = N^- \mu \left(e^{-\frac{\Delta E}{kT}} - 1 \right) \quad (1.3)$$

$$= N \mu \frac{e^{-\frac{\Delta E}{kT}} - 1}{e^{-\frac{\Delta E}{kT}} + 1} = N \mu \tanh \left(\frac{\Delta E}{2kT} \right) \quad (1.4)$$

$$\approx \frac{N \mu \Delta E}{2kT} = \frac{N \mu \gamma \hbar B}{2kT}. \quad (1.5)$$

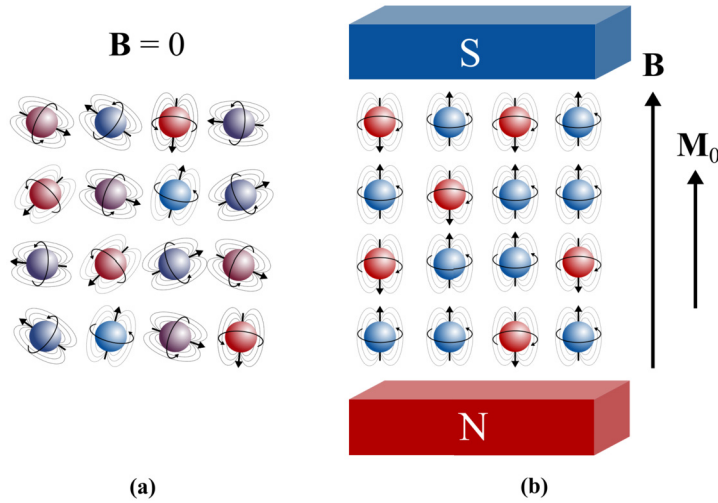
Here, $N^- = N / (e^{-\frac{\Delta E}{kT}} + 1)$, with $N = N^+ + N^-$, was used in (1.4), which can be easily derived from Eq. (1.2). Further, $\tanh(x) \approx x$ was approximated under the assumption of $x \ll 1$. Eq. (1.5) can be similarly derived for other spin values and is known as Curie's law. As illustrated in Fig. 1.3, this describes the first fundamental step in forming a magnetization that can later be used for imaging.

Principle 1: Net magnetization

If samples containing nuclei with nonzero spin s are brought into a magnetic field $\mathbf{B}$, a net-magnetization $\mathbf{M}_0$ forms. This magnetization is formed parallel to the external field ($\mathbf{M}_0 \parallel \mathbf{B}$), and its magnitude is characterized as

$$\mathbf{M}_0 = \frac{N \gamma s(s+1) \hbar^2}{3kT} \mathbf{B}, \quad (1.6)$$

with the gyromagnetic ratio γ , the spin quantum number s , the number of nuclei per volume N , the temperature T , and the Boltzmann constant k .

**FIGURE 1.3**

(a) Spins and corresponding magnetic moments are randomly oriented in the absence of an external magnetic field ($\mathbf{B} = 0$). (b) In the presence of an external magnetic field $\mathbf{B}$, however, the majority of the spins occupy the lower energy configuration, the spin-up state, generating a net magnetization $\mathbf{M}_0$ aligned with the magnetic field.

1.2.3 Magnetization dynamics

Forming a magnetization along the external magnetic field proves to be not very fruitful by itself. Instead, it helps to consider the magnetization dynamics over time. Any sample with a magnetic dipole moment $\mathbf{M}(t)$ experiences a torque $\mathbf{T}(t)$ in the presence of a static external field $\mathbf{B}$:

$$\mathbf{T}(t) = \mathbf{M}(t) \times \mathbf{B}. \quad (1.7)$$

Accordingly, the same happens to a sample with nonzero spin nuclei. However, as initially explained, the spin is also associated with an angular momentum. Thus, this torque affects the cumulative angular momentum $\mathbf{L}_s$ of the sample as²

$$\dot{\mathbf{L}}_s(t) = \mathbf{M}(t) \times \mathbf{B}. \quad (1.8)$$

The gyromagnetic ratio γ links the magnetic and the angular momentum of our spin ensemble. Hence, we can link the dynamics of the angular momentum to that of the net magnetization as presented in the following principle.

² For convenience, we use Newton's notation for differentiation throughout this chapter. If f is a function of t , then the derivative of f with respect to t is denoted as $\dot{f}$.

Principle 2: Larmor precession

The net magnetization $\mathbf{M}(t)$ described in Principle 1 undergoes the following dynamics in a static external magnetic field $\mathbf{B}$.³

$$\dot{\mathbf{M}}(t) = \gamma \mathbf{M}(t) \times \mathbf{B} + \text{relaxation terms}, \quad (1.9)$$

with the gyromagnetic ratio γ . These dynamics are known as Larmor precession [10].

This enables us to gain some important insights into the magnetization. First, if the magnetization is parallel to the external magnetic field, no dynamic behavior is observed

$$\mathbf{M}(t) \parallel \mathbf{B} \implies \mathbf{M}(t) \times \mathbf{B} = 0 \implies \dot{\mathbf{M}} = 0 \implies \mathbf{M}(t) = \mathbf{M}(0). \quad (1.10)$$

However, if the magnetization is skewed away from the external field by an angle θ , Principle 2 dictates that the gradient direction is tangential to a circle around the main magnetic field, and its magnitude is found as

$$\|\dot{\mathbf{M}}(t)\| = \gamma \sin(\theta) M B, \quad (1.11)$$

with the (static) magnitude of the magnetization $M = \|\mathbf{M}(t)\|$ and the field strength $B = \|\mathbf{B}\|$.

Thus, the magnetization vector describes a cone around the external magnetic field, with the tip lying on a circle. This type of motion is called precession and can again be likened to our spinning top. If the spinning top is nudged so that its rotational axis is angled with respect to the gravitational force, it precesses comparably around this axis.

The circle prescribed in this motion has radius $\sin(\theta)M$. Thus, the frequency of the rotation can be obtained as

$$\frac{\gamma \sin(\theta) M B}{\sin(\theta) M} = \gamma B = \omega. \quad (1.12)$$

This field-dependent angular frequency is also known as the *Larmor frequency* and will prove important for our understanding of the MRI signal.

1.3 NMR/MRI signal

1.3.1 Signal creation and reception

There is little use in knowing the magnetization dynamics if we cannot manipulate or detect them. We will now explore how the resonance phenomenon can be employed to change the magnetization and how this can be read out into a detectable signal.

³ The relevant relaxation terms will be introduced in Subsection 1.3.2.

1.3.1.1 Radiofrequency pulses

In MRI a signal can only be created if the magnetization is nutated away from the external magnetic field. For this purpose, the special case of rotating external magnetic fields is particularly relevant. To this end, we start by defining the overall magnetic field $\mathbf{B}_{tot}(t)$ as

$$\mathbf{B}_{tot}(t) = \begin{pmatrix} B_1 \cos(\omega_1 t) \\ B_1 \sin(\omega_1 t) \\ B_0 \end{pmatrix}, \quad (1.13)$$

with a stationary magnetic field with strength B_0 along the z -axis and a secondary magnetic field with amplitude B_1 rotating with angular frequency ω_1 in the transverse plane, i.e., the plane orthogonal to the $\mathbf{B}_0$ field.

To better understand the effects of rotating magnetic fields, it is useful to consider a rotating frame of reference (RFR). RFRs are a familiar concept. The reader is likely observing this chapter in an RFR: In a galactic reference frame, this book would rotate around the axis of the globe. However, the reference frame we observe is spinning with the globe and makes the book appear stationary. Similarly, we can define an RFR to rotate with the rotating magnetic field.⁴ For the stationary or laboratory frame of reference (x, y, z) and the RFR (x', y', z') , we can define the following reference frame transformation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} : \mathbf{R}(t) = \begin{pmatrix} \cos(\omega_1 t) & \sin(\omega_1 t) & 0 \\ -\sin(\omega_1 t) & \cos(\omega_1 t) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (1.14)$$

with $\mathbf{R}(t)$ being a proper rotation matrix, i.e., $\mathbf{R}(t)\mathbf{R}^T(t) = \mathbf{I}$ and $\det(\mathbf{R}(t)) = 1$.

Observing the magnetization dynamics of a magnetization $\mathbf{M}'(t) = \mathbf{R}(t)\mathbf{M}(t)$ in an RFR in the presence of a rotating dynamic magnetic field $\mathbf{B}_{tot}(t)$ such as defined in Eq. (1.13) yields

$$\dot{\mathbf{M}}'(t) = \frac{d}{dt} (\mathbf{R}(t)\mathbf{M}(t)) = \dot{\mathbf{R}}(t)\mathbf{M}(t) + \mathbf{R}(t)\dot{\mathbf{M}}(t) \quad (1.15)$$

$$= \dot{\mathbf{R}}(t)\mathbf{R}^T(t)\mathbf{M}'(t) + \mathbf{R}(t)\gamma\mathbf{M}(t) \times \mathbf{B}_{tot}(t). \quad (1.16)$$

By multiplying out the matrices, it can quickly be concluded that

$$\dot{\mathbf{R}}(t)\mathbf{R}^T(t)\mathbf{M}'(t) = -\omega_1 \mathbf{M}'(t) \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (1.17)$$

and

$$\mathbf{M}(t) \times \mathbf{B}(t) = \mathbf{R}^T(t)\mathbf{M}'(t) \times \mathbf{B}_{tot}(t), \quad (1.18)$$

⁴ An extensive explanation of the RFR systems in MRI can be found in [33].

due to the rotation properties of the cross product. It follows that

$$\dot{\mathbf{M}}'(t) = -\omega_1 \mathbf{M}'(t) \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \mathbf{R}(t) \gamma \mathbf{R}^T(t) \mathbf{M}'(t) \times \mathbf{B}_{tot}(t) \quad (1.19)$$

$$= \gamma \left(\mathbf{B}'_{tot}(t) - \begin{pmatrix} 0 \\ 0 \\ B_0 \frac{\omega_1}{\omega_0} \end{pmatrix} \right), \quad (1.20)$$

with the Larmor frequency $\omega_0 = \gamma B_0$ and

$$\mathbf{B}'_{tot}(t) = \mathbf{R}(t) \mathbf{B}(t) = \begin{pmatrix} B_1 \\ 0 \\ B_0 \end{pmatrix}. \quad (1.21)$$

This yields the following principle:

Principle 3: Rotating frame excitation

For a magnetic field with a rotational component in the xy -plane $\mathbf{B}_{tot}(t) = (B_1 \cos(\omega_1 t) \quad B_1 \sin(\omega_1 t) \quad B_0)^T$, the net magnetization $\mathbf{M}'(t)$ in a RFR rotating about the z -axis with frequency ω_1 can be described as

$$\dot{\mathbf{M}}'(t) = \gamma \mathbf{M}'(t) \times \mathbf{B}_{eff}(t) + \text{relaxation terms}, \quad (1.22)$$

with the gyromagnetic ratio γ , the Larmor frequency $\omega_0 = \gamma B_0$ and

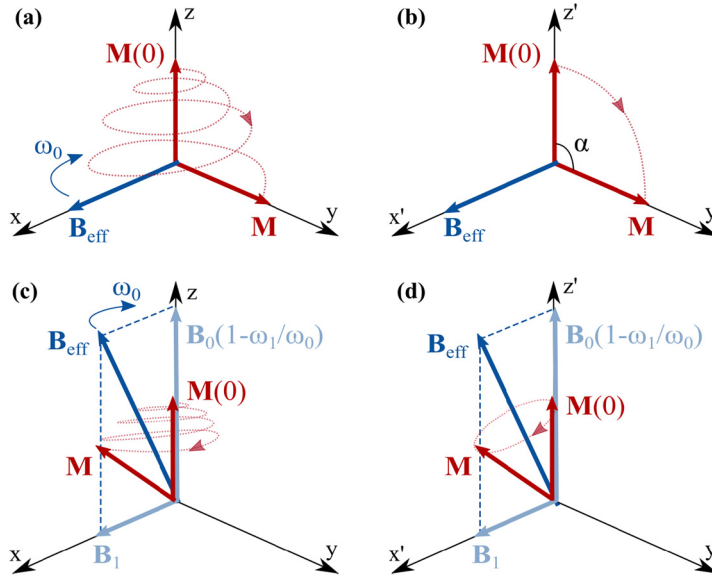
$$\mathbf{B}_{eff}(t) = \begin{pmatrix} B_1 \\ 0 \\ B_0 \left(1 - \frac{\omega_1}{\omega_0}\right) \end{pmatrix}. \quad (1.23)$$

Again, this principle allows us to derive some important properties. Namely, if the secondary magnetic field $\mathbf{B}_1$ is applied at the resonance frequency, i.e., $\omega_1 = \omega_0$, the magnetization vector in the RFR rotates about the secondary magnetic field only, as shown in Fig. 1.4. The speed of this rotation is determined by the strength of the secondary magnetic field. Hence, the angle accumulated by the secondary magnetic field of time-varying amplitude $B_1(t)$ can be computed as

$$\alpha = \int_0^\infty \gamma B_1(t) dt, \quad (1.24)$$

and is called the *flip angle*. Accordingly, for the simple case of a constant magnetic field B_1 applied during a time τ , the flip angle $\alpha = \gamma B_1 \tau$. A flip angle of 90° tips the magnetization into the transverse plane and yields the strongest magnetization component orthogonal to the main magnetic field. The application of $\alpha = 180^\circ$ returns no transverse magnetization, essentially inverting the magnetization.

On the other hand, for $\omega_1 \neq \omega_0$ the magnetization precesses around an angled effective field, where the angle between $\mathbf{B}_{eff}$ and the z -axis amounts to $\theta = \text{atan}(B_1 / (B_0 (1 - \omega_1 / \omega_0)))$. In MRI, the field

**FIGURE 1.4**

On-resonance 90° excitation pulse ($\mathbf{B}_{eff} = \mathbf{B}_1$, $\omega_1 = \omega_0$) tipping the magnetization to the transverse plane, as viewed in the (a) laboratory and (b) rotating frame of reference (RFR). (c, d) Off-resonance excitation pulse ($\omega_1 \neq \omega_0$), causing $\mathbf{B}_{eff}$ to be angled towards the z -axis, preventing full excitation.

strength of the secondary magnetic field is much smaller than the main magnetic field ($B_1 \ll B_0$), thus, to induce a precession that rotates the magnetization away from the z -axis, it is important to closely match the resonance frequency.

In clinical MRI systems, the field strength is often around $B_0 = 1.5T$. At this field strength, resonance is achieved at the Larmor frequency of $\omega_0/(2\pi) \approx 63.9 \text{ MHz}$ (for the hydrogen nucleus). As these frequencies are in the range of radio waves, the rotating magnetic field $\mathbf{B}_1$ may be referred to as the radiofrequency (RF) field. Because the B_1 field is employed for a short duration, it is also often commonly referred to as an RF pulse. An RF pulse creating a flip angle of 90° is (conveniently) called a 90° pulse.

1.3.1.2 Signal detection

To better understand how to create a detectable signal from the magnetization $\mathbf{M}$ in MRI, it is useful to consider the individual magnetization components

$$\mathbf{M}(t) = \begin{pmatrix} M_x(t) \\ M_y(t) \\ M_z(t) \end{pmatrix}. \quad (1.25)$$

$M_z(t)$ describes the magnetization component along the z -axis that is aligned with the external magnetic field $\mathbf{B}_0 = (0 \ 0 \ B_0)^T$, and is called the longitudinal magnetization. The vector $\mathbf{M}_{xy} = (M_x \ M_y)^T$ is referred to as the transverse magnetization that lies in the xy -plane, orthogonal to the main magnetic field $\mathbf{B}_0$.

Using this notation, the net magnetization $\mathbf{M}$ that is formed, following Principle 1, is aligned with the external magnetic field $\mathbf{B}_0$

$$\mathbf{M}(0) = \begin{pmatrix} 0 \\ 0 \\ M_0 \end{pmatrix}. \quad (1.26)$$

Using Principle 3, we can calculate the effect of an RF pulse played along the y' -axis with a duration τ and strength B_1 chosen to achieve a flip angle $\alpha = \gamma \tau B_1 = 90^\circ$. When neglecting the relaxation terms, it follows for the magnetization that

$$\mathbf{M}'(\tau) = \begin{pmatrix} M_0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \mathbf{M}(\tau + t) = \begin{pmatrix} \cos(\omega_0 t) M_0 \\ \sin(\omega_0 t) M_0 \\ 0 \end{pmatrix}, \quad (1.27)$$

in the RFR and the stationary frame, respectively.

This means that a time-varying magnetization component has been created in the transverse plane. The time-varying magnetic field originating in the precession of the magnetization vectors can be picked up using an inductive coil near the sample. In its most simplistic form, this coil is an individual loop of wire. Faraday's law of induction declares that a time-varying magnetic flux through a wire loop produces a current in the loop and induces a voltage ϕ over the edges of the loop.

Principle 4: Faraday's law of induction

If a coil is placed such that a magnetic field has a temporally varying component normal to the coil area, a voltage is induced in the coil. The produced voltage E is proportional to the negative of the rate of temporal variation of the flux ϕ that in turn is proportional to the magnetic field strength normal to the coil ($\phi \propto \mathbf{M}_{xy}$):

$$E \propto -\frac{d\phi}{dt}. \quad (1.28)$$

The demand for a time-varying magnetic flux is the reason solely precessing magnetization in the xy -plane yields an increase to an NMR signal: the z -component does not precess and, hence, generates no voltage. At greater strengths of the $\mathbf{B}_0$ field, the magnetization precesses at a higher frequency, and so the value of $d\phi/dt$ rises. Thus, stronger magnetic fields yield better signal strength not only because of the larger nuclear polarization (Principle 1) but also because of the additional increase in magnetic flux.

1.3.2 Signal relaxation and decay

When researchers first created NMR signals from the principles just cited, such as Erwin Hahn with his pulsed NMR experiments in 1950 [8], they made the sobering observation that the carefully cre-

ated signal would quickly decay. The signal that was induced in a nearby coil in this experiment was termed free induction decay (FID). It showed an oscillating behavior at the Larmor frequency, but also decay of the amplitude of those oscillations was observed. It was Felix Bloch who phenomenologically concluded that the created NMR signal is subject to multiple relaxation and decay processes, which prevent a transverse magnetization from indefinite signal induction [2].

To create a detectable signal, following Principle 3, an RF pulse needs to be applied at the Larmor frequency. Note that this frequency corresponds to the energy gap between the spin states

$$\Delta E = \omega_0 h = \gamma h \mathbf{B}_0, \quad (1.29)$$

where h is the Planck constant. Thus, when a secondary magnetic field is applied, energy is transmitted into the system. This excess energy excites the system above the thermal equilibrium, to which it naturally relaxes back. In the vector model, the M_z component has been lowered from its equilibrium value of M_0 , and the M_x and/or M_y components may have a nonzero value. Each of the magnetization components M_z , M_x , and M_y returns to the thermal equilibrium value over time. To better understand this relaxation, it is useful to consider two different processes: 1) The recovery of the M_z component to the thermal equilibrium value M_0 , also known as longitudinal relaxation, and 2) The decay of the transverse magnetization towards zero, also known as transverse relaxation.

1.3.2.1 Longitudinal relaxation

During the recovery to thermal equilibrium, the spin system dissipates energy to the surrounding lattice. Thus, the recovery of the longitudinal magnetization is also known as spin-lattice relaxation. The closer the system gets to thermal equilibrium, the closer the longitudinal magnetization is to the equilibrium magnetization M_0 . The rate of recovery depends on the excess of energy in the spin system, as well as a sample-specific constant T_1 . This allows us to describe the dynamics of longitudinal magnetization, including magnetization recovery, as

$$\dot{\mathbf{M}}_z(t) = \gamma (\mathbf{M}(t) \times \mathbf{B}(t))_z - \frac{M_z(t) - M_0}{T_1}, \quad (1.30)$$

where $(\mathbf{x})_z = x_z$, for $\mathbf{x} = (x_x \ x_y \ x_z)^T$.

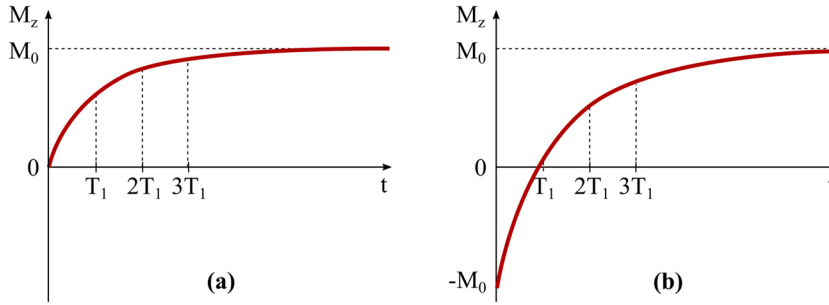
In MRI, observing the longitudinal magnetization following an RF pulse of flip angle α is of particular interest. Following Principle 3, we see that, promptly after the application of the RF pulse, the longitudinal magnetization M_z can be described by $M_0 \cos(\alpha)$. For a time t after the RF pulse, Eq. (1.30) gives the value of M_z as

$$M_z(t) = M_0 \cos(\alpha) + (M_0 - M_0 \cos(\alpha)) \left(1 - e^{-\frac{t}{T_1}}\right). \quad (1.31)$$

For instance, following a 90° pulse, the value of M_z is given by

$$M_z(t) = M_0 \left(1 - e^{-\frac{t}{T_1}}\right). \quad (1.32)$$

This recovery and an example of longitudinal magnetization regrowth following a 180° pulse can be seen in Fig. 1.5.

**FIGURE 1.5**

Exponential regrowth, over time, of the longitudinal component of the magnetization $M_z(t)$ to the equilibrium value M_0 after a 90° (a) or 180° (b) excitation pulse.

1.3.2.2 Transverse relaxation

The physical basis for the second relaxation mechanism, the transverse magnetization decay, differs from the T_1 relaxation process. A coherent magnetization in the transverse plane can be maintained only as long as all magnetic moments contributing to the transverse magnetization precess at the same frequency. However, even in a perfectly homogeneous magnetic field, magnetic moments precess at marginally different frequencies. This is due to the magnetic moments of neighboring nuclei that influence the observed net magnetic field strength at any location and in a time-varying manner, due to the molecular motion. Transverse magnetization decay is caused by losing the *phase coherence* among the magnetic moments. As a result, the net magnetization decays over time (Fig. 1.6): The magnetic moments fan out, and consequently, the magnitude of the net sum decreases. Thus, the transverse magnetization unwinds back to the thermal equilibrium value of zero with a sample-specific time constant called the spin–spin relaxation time, or T_2 :

$$\dot{M}_x(t) = \gamma(\mathbf{M}(t) \times \mathbf{B}(t))_x - \frac{M_x(t)}{T_2}, \quad (1.33)$$

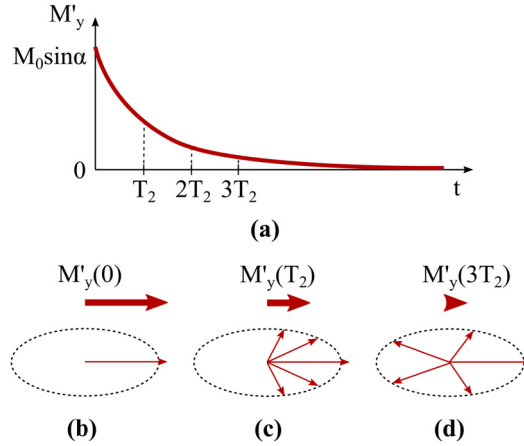
$$\dot{M}_y(t) = \gamma(\mathbf{M}(t) \times \mathbf{B}(t))_y - \frac{M_y(t)}{T_2}. \quad (1.34)$$

Together these relaxation processes constitute the Bloch equations, one of the most important tools for MRI researchers.

Principle 5: Bloch equations

The time evolution of M_z , M_x , and M_y can be described by differential equations, recognized as the Bloch equations:

$$\dot{\mathbf{M}}(t) = \frac{d}{dt} \begin{pmatrix} M_x \\ M_y \\ M_z \end{pmatrix} = \gamma(\mathbf{M}(t) \times \mathbf{B}(t)) - \begin{pmatrix} \frac{M_x(t)}{T_2} \\ \frac{M_y(t)}{T_2} \\ \frac{M_z(t) - M_0}{T_1} \end{pmatrix} \quad (1.35)$$

**FIGURE 1.6**

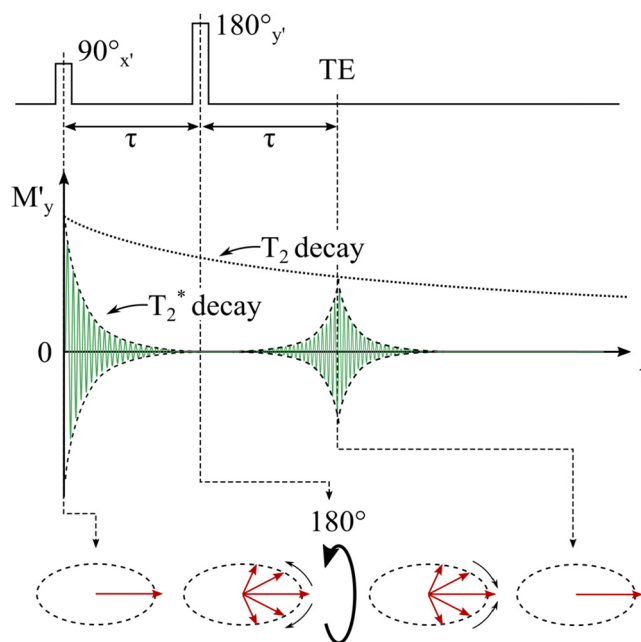
(a) Exponential decay of the transverse component of the magnetization ($M'_y(t)$) in the RFR over time. The transverse component recovers to the equilibrium value zero after excitation with an arbitrary flip angle α . The coherence loss of magnetic moments precessing in the transverse plane and the corresponding net transverse magnetization M_y is shown right after the excitation pulse (b) and after one (c) or several (d) spin-spin relaxation periods T_2 .

Two mechanisms that contribute to the loss of phase coherence of the transverse magnetization can be differentiated. The first mechanism is the “real” T_2 decay originating in spin-spin interaction, as described before. The second mechanism derives from the spatial disparities of the magnetic field strength inside the body or across the sample. There are, in turn, two notable causes for these differences. The first is the inherent magnet design: It is impracticable to design a magnet generating a perfectly homogeneous magnetic field throughout the whole subject. The second cause is given by regional changes in the magnetic field due to the distinct magnetic susceptibilities of different tissues. This effect is particularly pronounced at the air/tissue and bone/tissue boundaries. Collectively, these effects create a decline in phase coherence, which is described by a relaxation time T_2' . The effective relaxation time that characterizes the decay of transverse magnetization is a combination of signal loss due to T_2 and T_2' effects. This is defined by T_2^* , and it can be expressed by

$$\frac{1}{T_2^*} = \frac{1}{T_2'} + \frac{1}{T_2}. \quad (1.36)$$

Considering these decay mechanisms, the detectable magnetization in the transverse plane from Eq. (1.27), which is formed after the application of a 90° RF pulse, can be described as

$$\mathbf{M}(t) = \begin{pmatrix} \cos(\omega_0 t) \\ \sin(\omega_0 t) \\ 0 \end{pmatrix} M_0 e^{-\frac{t}{T_2^*}}. \quad (1.37)$$

**FIGURE 1.7**

Signal evolution during a spin-echo experiment in the RFR (green, solid) and the laboratory frame of reference (gray, dashed). After the first 90° pulse, spins start dephasing, generating the FID signal, characterized by the T_2^* decay. The application of a 180° pulse inverts the process and generates an echo at $TE = 2\tau$, recovering T_2' losses. Overall, the spin-echo signal follows a T_2 decay.

In high-resolution NMR spectroscopy for chemical examination, the sample is small and spatially uniform. Thus, there are only small field inhomogeneities leading to long T_2' times. In this case, the value of T_2^* is adequately approximated by T_2 , and Principle 5 is reasonably accurate for representing the decay of transverse magnetization in the entire sample. In MRI, however, the value of T_2' can be up to 10–100 times lower than that of T_2 . Thus, T_2' is often the main driver of T_2^* decay.

In practice, it can be important to discriminate between T_2 and T_2^* . The field variations contributing to T_2 are temporally variant, but the effects contributing to T_2^* are static over time. Fortunately, a mechanism can also be used to recover the signal loss due to temporally invariant causes. If, at a time τ , following the 90° pulse, an additional 180° pulse is applied, the magnetization is flipped. As shown in Fig. 1.7, phase dispersion that has been caused by temporally invariant inhomogeneities, is then rewound. Thus, this pulse is also known as *refocusing pulse*. At the time point *echo time* ($TE = 2\tau$), when the phase coherence is maximally restored, a *spin-echo* is formed, which compensated for T_2' decay, and T_2 can be measured [8][22].

1.4 Image formation

The NMR signal defined so far is the gross summation of all net magnetic moments in the sample. It was not before 1973 when the contributions of Paul C. Lauterbur enabled to make spatial sense of the signal [11]. The key innovation was the introduction of *gradient fields*, magnetic fields whose strength is a function of the spatial location. It was discovered that, in the presence of a gradient field, the spatial distribution of the protons gives rise to a variety of proton resonant frequencies, each dependent upon the location of the proton in the sample.

To understand image formation using magnetic field gradients, consider the most common form, linear field gradients. Let $\mathbf{G} = \nabla \mathbf{B}$ denote the *gradient* of the magnetic field strength of this gradient field. The total magnetic field strength at a position $\mathbf{r}$ can then be obtained as

$$B(\mathbf{r}) = B_0 + \mathbf{G}^T \mathbf{r}. \quad (1.38)$$

It should be noted that the orientation of the gradient fields (but not the direction of change) is always parallel to the static field, in order to alter the longitudinal component of the overall magnetic field.

In the presence of a field gradient, the Larmor frequency of the magnetization becomes a function of its spatial location,

$$\omega(\mathbf{r}) = \gamma \mathbf{G}^T \mathbf{r} \quad (1.39)$$

In other words, the gradient fields allow linking the Larmor frequency to the spatial location inside the magnet.

To see how the difference in precession frequency can be used for encoding, consider the application of gradient fields after a transverse magnetization has been created. For this purpose, the transverse magnetization is best described as a complex function of the time and the location $M_{xy}(t, \mathbf{r}) = M_x(t, \mathbf{r}) + i M_y(t, \mathbf{r})$. With this definition, the complex plane of $M_{xy}(t, \mathbf{r})$ coincides with the xy -plane. Following Principle 2, and assuming no relaxation effects due to $t \ll T_2^*$, it is easily verified that the precession can be described as

$$M_{xy}(t, \mathbf{r}) = e^{-i\gamma B(\mathbf{r})t} M_{xy}^0(\mathbf{r}). \quad (1.40)$$

Combining Eqs. (1.40) and (1.38), yields

$$M_{xy}(t, \mathbf{r}) = e^{-i\gamma B_0 t} e^{-i\gamma \mathbf{G}^T \mathbf{r} t} M_{xy}^0(\mathbf{r}). \quad (1.41)$$

The MRI signal can then be obtained as the spatial sum across the imaging volume

$$S'(t) \propto \int e^{-i\gamma B_0 t} e^{-i\gamma \mathbf{G}^T \mathbf{r} t} M_{xy}^0(\mathbf{r}) d\mathbf{r} \quad (1.42)$$

$$= e^{-i\omega_0 t} \int e^{-i\gamma \mathbf{G}^T \mathbf{r} t} M_{xy}^0(\mathbf{r}) d\mathbf{r}. \quad (1.43)$$

After demodulation of the signal with the Larmor frequency ω_0 ,

$$S(t) \propto \int e^{-i\gamma \mathbf{G}^T \mathbf{r} t} M_{xy}^0(\mathbf{r}) d\mathbf{r}. \quad (1.44)$$

The well-versed reader might recognize the resemblance to a Fourier transform. Indeed, to connect to the Fourier transform, consider a gradient $\mathbf{G} = (G_x \ G_y \ G_z)^T$ after time t . The gradient moments can be defined as $m_x = G_x t$, $m_y = G_y t$ and $m_z = G_z t$ and wavenumbers as $k_x = \gamma m_x$, $k_y = \gamma m_y$ and $k_z = \gamma m_z$ to yield

$$S(k_x, k_y, k_z) \propto \int \int \int e^{-i(k_x x + k_y y + k_z z)} M_{xy}^0(x, y, z) dx dy dz. \quad (1.45)$$

It follows that the Fourier transform of the signal S recovers the spatial information of the magnetization.

Principle 6: Gradient encoding

If gradient fields with gradient moments m_x , m_y and m_z are applied after the creation of a transverse magnetization $M_{xy}^0(x, y, z)$, the resulting signal can be described as

$$S(k_x, k_y, k_z) \propto \int \int \int e^{-i(k_x x + k_y y + k_z z)} M_{xy}(x, y, z) dx dy dz, \quad (1.46)$$

using the wavenumbers $k_x = \gamma m_x$, $k_y = \gamma m_y$, and $k_z = \gamma m_z$, and the gyromagnetic ratio γ .

Of note, the Fourier transform of this signal recovers the spatial information about the magnetization

$$\mathbf{F}(S)(x, y, z) \propto M_{xy}^0(x, y, z). \quad (1.47)$$

In the remainder of this section it will be explored, how this principle can be used to encode an image in MRI. Commonly, three steps can be differentiated in this process: frequency encoding, phase encoding, and slice selection. Principle 6 is commonly considered in a stationary reference frame aligned with the image orientation to ease notation. Thus, we can assume that frequency encoding is performed along the x -axis, phase encoding along the y -axis, and slice selection along the z -axis.

1.4.1 Frequency encoding

Frequency encoding is the most straightforward way to apply Principle 6 for spatial encoding. In frequency encoding, a gradient G_x is turned on and the signal is continuously sampled, while the gradient is applied. The acquired signal can thus be described as

$$S(\gamma G_x t, 0, 0) \propto \int e^{-i k_x x} \int \int M_{xy}(x, y, z) dy dz dx. \quad (1.48)$$

Accordingly, one spatial dimension can be fully characterized using frequency encoding.

Since the frequency encoding gradient G_x remains turned on, the spins precess at different frequencies, while the signal is acquired. Thus, the received composite signal is a summation of the contributions at many frequencies. This situation can be compared to a glass harp (Fig. 1.8a). Each spin is a different wine glass, and the filling line that determines the frequency is a function of the position along the frequency encoding direction. In this analogy, frequency encoding is obtained by playing all glasses at the same time, and a frequency analysis can be used to recover the position of the individual glasses.

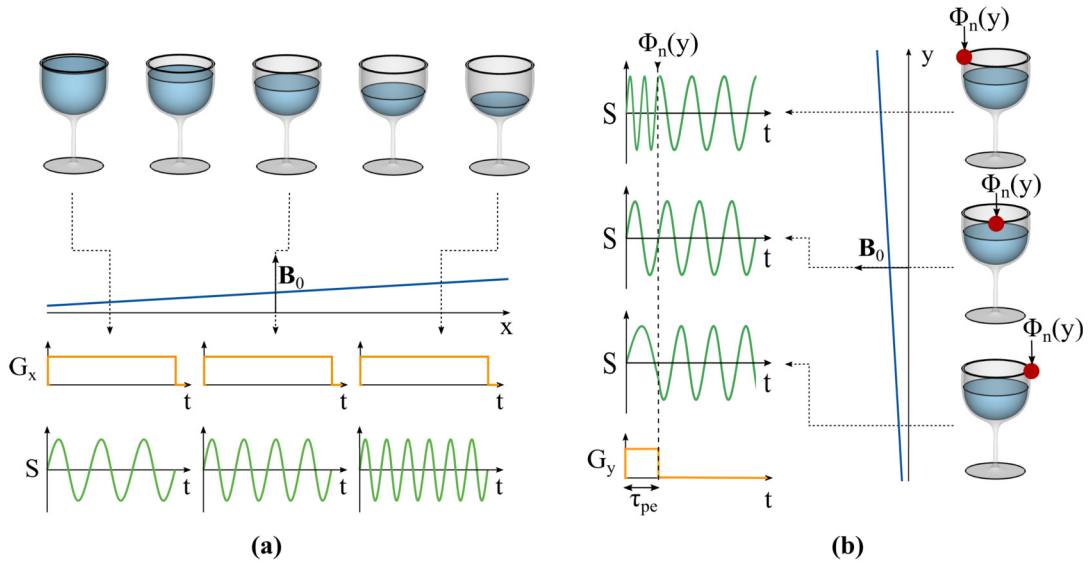


FIGURE 1.8

(a) Frequency encoding: when G_x is applied, it induces a spatially dependent shift in the B_0 field and causes spins, at different locations along the x -axis, to precess at different frequencies. This enables signal localization in the x direction, similar to playing glass harp, with wine glasses filled to different water levels, where the higher the water level, the lower the pitch. (b) Phase encoding: When G_y is applied for a duration τ_{pe} , the spins accumulate a spatially dependent phase value ϕ , precessing with the same frequency but different phases once G_y is turned off. This can be compared to partially water-filled glasses, with identical water levels, being played at different initial phases.

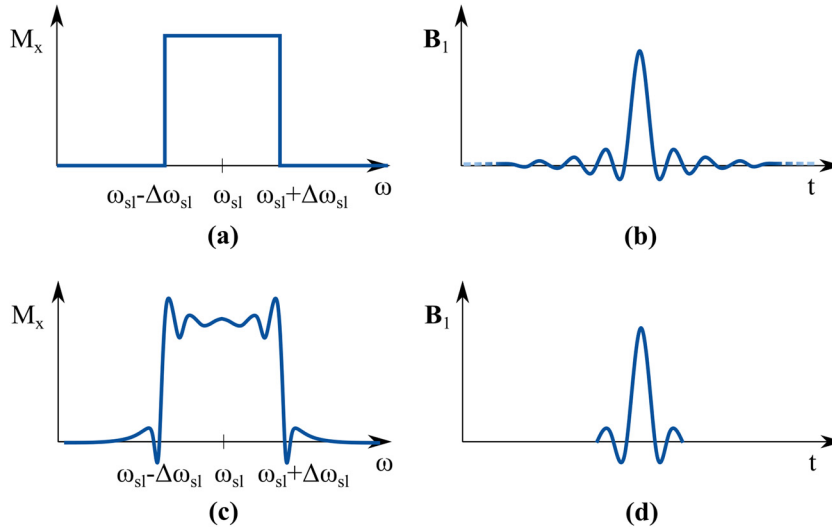
1.4.2 Phase encoding

In order to encode additional spatial dimensions, phase encoding can be applied. In phase encoding, a gradient is turned on and off *before* the frequency encoding and data acquisition begins. A number N_y of distinct values of this phase encoding gradient with stepwise amplitude increment ΔG_y are used for a fixed duration τ_{pe} , to obtain different gradient moments $m_y(n) = n \Delta G_y \tau_{pe}$, $-N_y < n \leq N_y$.

In combination with frequency encoding, we obtain the following signal:

$$S(\gamma G_x t, \gamma n \Delta G_y \tau_{pe}, 0) \propto \int \int e^{-ik_x x + k_y y} \int M_{xy}^0(x, y, z) dz dy dx. \quad (1.49)$$

During the application of G_y with moment $m_y(n)$, spins accumulate a spatially dependent phase shift $\phi_n(y) = \gamma m_y(n) y$. So, while all spins along the phase encode dimension precess at the same frequency, the phase between the signals is shifted. Since only one phase shift can be obtained in the phase encoding direction, the application of many gradient amplitudes is necessary to spatially resolve this information. In the glass-harp analogy, phase encoding is represented by glasses with the same

**FIGURE 1.9**

(a) Ideal slice profile of an excitation RF pulse centered at frequency ω_{sl} , with an excitation bandwidth $\pm\Delta\omega_{sl}$, and (b) its corresponding temporal profile, an infinitely long sinc pulse. (d) Example of a time-limited RF pulse, obtained by truncating a sinc pulse with a rectangular window. This causes ripples at the edge of the slice profile, the so-called Gibbs ringing (c).

filling line, thus producing the same frequency, but they are played with a different phase, as illustrated in Fig. 1.8b.

Phase encoding, as just described, can be applied along multiple dimensions. For example, additional phase encoding can be used in the slice dimension to enable the acquisition of a 3D image. To this end, a second phase encoding gradient is applied such that, for each gradient step in the new axis, a whole set of phase and frequency encoding steps are obtained. However, the number of phase encoding steps required grows exponentially with the number of dimensions encoded. Because phase encoding is the major time-limiting factor in MRI, 3D imaging is often time-consuming, and the acquisition of (multiple) 2D slices remains standard in clinical use.

1.4.3 Slice selection

To acquire a 2D slice in MRI, slice selection can be accomplished by a frequency-selective RF pulse employed together with a magnetic field gradient G_z . During the application of G_z , the Larmor frequency varies as a function of the location along the slice-encoding direction. Thus, if the selective RF pulse is administered at a frequency ω_{sl} with an excitation bandwidth of $\pm\Delta\omega_{sl}$ (Fig. 1.9), then following Principle 3 only protons precessing at frequencies between $\omega_{sl} + \Delta\omega_{sl}$ and $\omega_{sl} - \Delta\omega_{sl}$ are flipped into the transverse plane; those with resonant frequencies outside this range are not effectively nutated and remain largely aligned with the z -direction. In combination with frequency and phase encoding,

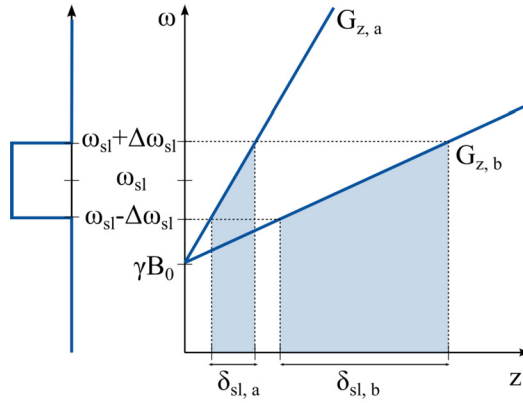
**FIGURE 1.10**

Illustration of two different gradient strengths, $G_{z,a} > G_{z,b}$, and their effect on slice selection. Considering an RF pulse with an excitation bandwidth of $2\Delta\omega_{sl}$, the slice thickness δ_{sl} can be varied by changing the gradient strength G_z .

slice selection enables two-dimensional imaging of the slice of interest, as follows:

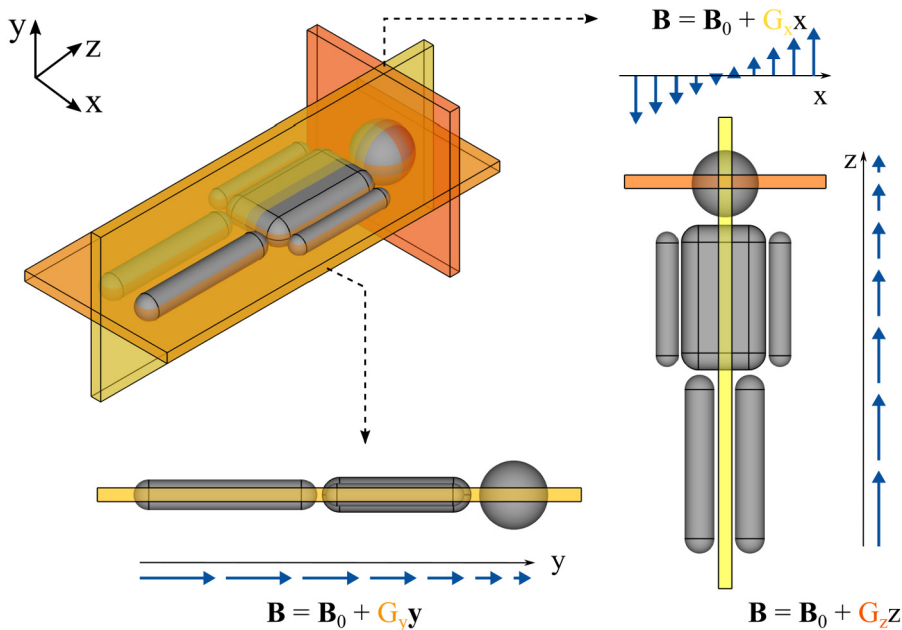
$$S(\gamma G_x t, \gamma n \Delta G_y \tau_{pe}) \propto \int_{slice} \int \int e^{-ik_x x + k_y y} M_{xy}^0(x, y, z) dx dy dz. \quad (1.50)$$

The thickness δ_{sl} of the acquired slice can be characterized by the bandwidth of the RF pulse, $2\Delta\omega_{sl}$, and the value of the slice-selection gradient:

$$\delta_{sl} = \frac{2\Delta\omega_{sl}}{\gamma G_z}. \quad (1.51)$$

The slice thickness can thus be increased either by reducing the strength of G_z or by broadening the frequency bandwidth of the RF pulse, as illustrated in Fig. 1.10. The ideal slice-excitation profile of an RF pulse would cause the magnetization to be excited with the target flip angle inside the slice thickness and with a zero flip angle outside. This corresponds to a rectangular slice profile. For small flip angles, the achieved slice profile can be approximated by the frequency spectrum of the RF pulse. Thus, the commonly chosen RF pulse shapes are an approximation to sinc pulses, whose frequency spectrum is rectangular (Fig. 1.9a–b). However, to obtain a perfectly rectangular slice profile, an infinitely long RF pulse would be required. In practice, truncated sinc pulses are employed (Fig. 1.9c–d). For a fixed pulse shape, increasing the pulse duration results in a narrowing of the frequency bandwidth and, consequently, smaller slice thickness. Shifting the center frequency ω_{sl} of the RF pulse, on the other hand, alters the slice position. Thus, the excitation pulse properties can be used to select the volume of interest.

The application of a slice-selection gradient also induces a shift of Larmor frequency within the excited slice. Hence, after slice selection, the phase is spatially inhomogeneous across the slice thickness, which results in signal loss. To reduce this undesired loss of phase coherence, a rephasing gradient

**FIGURE 1.11**

Sagittal, coronal, and axial slices acquired by choosing the slice-selection direction along the x , y , or z -axis (in the laboratory frame), respectively. Gradients induce spatially dependent shifts in the magnetic field strength B_0 along the direction of their application, allowing the excitation of narrow bands of spins in this direction.

with opposite polarity is employed for half the duration to match the gradient moment. The area of the rephasing lobe is half of the slice selection gradient lobe since dephasing only affects the transverse magnetization M_{xy} . A viable approximation is to assume that, during the RF excitation, the magnetization is instantaneously tipped into the transverse plane at the center of the pulse, thus dephasing only occurs during the second half of the RF pulse duration.

1.4.4 Sequence diagram

So far, all encoding steps were considered in the image reference system. However, if frequency, phase, or slice encoding is angled with respect to the laboratory frame, multiple physical gradient coils are employed for each of the encoding steps. The three standard orientations, sagittal, coronal, and axial (Fig. 1.11) can be obtained by performing slice selection along with the laboratory frame x , y , or z gradient axis, respectively. To acquire an oblique slice, two or three of the gradients are employed concurrently with suitably weighted strengths to achieve slice selection.

Fig. 1.12 depicts a simple MRI pulse sequence diagram. At this point, the reader should be equipped with all the tools required to interpret the sequence diagrams. The imaging process starts with the application of an RF pulse with a flip angle α . The shape of the RF pulse is chosen as a truncated sinc, as illustrated in Fig. 1.9. At the same time, a gradient for slice selection G_z is applied, followed by the

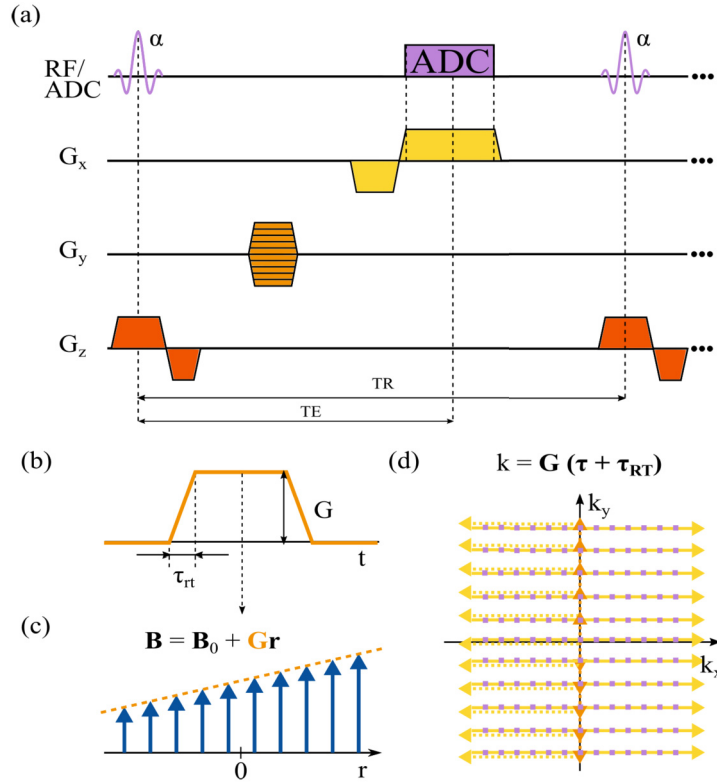


FIGURE 1.12

(a) Pulse sequence diagram example, showing an α -RF excitation pulse and a simultaneous slice-selection gradient with subsequent refocusing lobe, followed by a phase-encoding step and, finally, the frequency-encoding gradient applied during signal acquisition. (b) Trapezoidal gradient shape, highlighting rise time τ_{rt} . (c) B_0 dependence on spatial location, with gradient strength determining the field-intensity slope. (d) k -space trajectory of the pulse sequence. Every time the pulse sequence snippet in a) is executed, a single line in k -space is acquired (yellow (light gray in print version)). Multiple repetitions with different phase encoding steps (orange (mid gray in print version)) enable the sampling of the 2D k -space in parallel lines. This sampling trajectory in k -space is known as Cartesian sampling.

slice rewinder. Next, the phase-encoding gradient G_y is played with one of many stepwise incremented amplitudes. To readout the frequencies along one line, a prewinder is played before the frequency encoding, such that, during frequency encoding, k_x is sampled from $-k_{x,max}$ to $k_{x,max}$. Finally, the frequency encoding gradient G_x is applied concurrently with the image readout, where the analog-to-digital converter (ADC) is switched on and the signal induced by the transverse magnetization M_{xy} is captured. As denoted in the sequence diagram, the most basic form of gradient field has a trapezoidal shape over time. The duration of the ramp is called rise time τ_{rt} that describes how long it takes for the gradient field to be ramped up to its maximum value G , as shown in Fig. 1.12b.

When characterizing pulse sequences, two timing parameters are particularly useful: 1) The duration between the creation of the transverse magnetization and the time point when the sampling along the frequency encoding direction accumulates zero moment from the pre- and rewinding. This time is commonly known as echo time or TE; 2) the time between the application of two excitation RF pulses in the same slice selection. This time is known as repetition time or TR. TE and TR are illustrated in Fig. 1.12a.

1.4.5 k -space formalism

Due to the naming of the wavenumbers k_x , k_y , and k_z , MRI is often considered to collect image data in “ k -space”. Following Principle 6 it is easily seen that k -space is indeed the Fourier domain of the MR image. The concept of k -space formalism is thoroughly discussed in [32,35].

For 2D imaging, in practical terms, k -space is a data matrix of size $N_r \times N_{pe}$. It can be derived from the properties of the Fourier transform that the distance between the k -space points determines the field-of-view (FOV) of the image, which describes the extent of the object to be imaged:

$$FOV_x = \frac{1}{\Delta k_x} \quad (1.52)$$

$$FOV_y = \frac{1}{\Delta k_y}. \quad (1.53)$$

In a dual principle, the extent to which k -space is sampled determines the image resolution

$$\Delta x = \frac{1}{2k_{x,max}} \quad (1.54)$$

$$\Delta y = \frac{1}{2k_{y,max}}. \quad (1.55)$$

1.4.6 k -space trajectories

The k -space can be traversed in multiple ways to acquire sufficient data for image reconstruction. k -space trajectory refers to the path along which this data is acquired. Most imaging sequences in clinical MRI sample k -space using a *Cartesian* sampling scheme (Fig. 1.12d and Fig. 1.13b). Here, a rectangular grid with equidistant k -space sampling in each of the two directions (k_x and k_y) is filled. In Cartesian sampling, one row of k -space is acquired at a time using frequency encoding, before navigating to the next row employing phase encoding. By collecting data in this fashion, high and low spatial frequencies are sampled uniformly. After the acquisition of all rows required to achieve the desired FOV and resolution, the k -space can be converted to an image using the Fourier transform. This will be discussed in further detail in Chapter 2 of this book.

1.4.6.1 Echo-planar imaging

A particular example of Cartesian k -space sampling is the so-called echo-planar imaging (EPI) [12]. In EPI, the collection of all the data necessary to reconstruct an image is performed after a single excitation of the magnetization in one set of echoes, commonly referred to as an echo train. Following the creation of the transverse magnetization, multiple k -space lines are sampled using frequency encoding. These

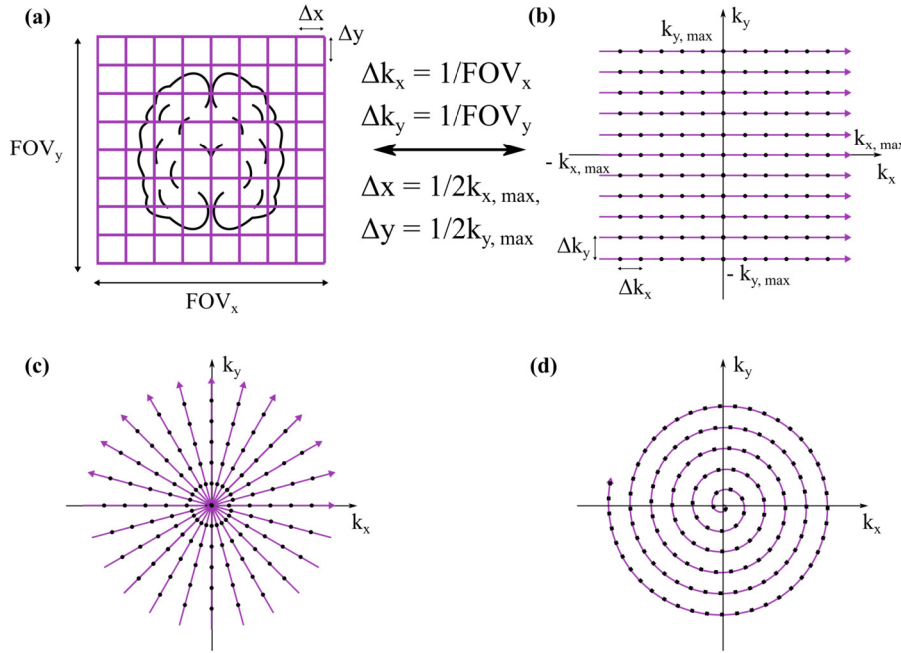
**FIGURE 1.13**

Image-space representation, illustrating the concepts of resolution (Δx , Δy) and field of view (FOV) (a) and their relationship with k -space for the example of Cartesian sampling (b). Examples of radial (c) and spiral (d) trajectories for k -space sampling.

are separated by a short phase-encoding gradient pulse to traverse along the phase-encoding direction. To allow for symmetric k -space readouts, this echo train is preceded by frequency- and phase-encoding prewinders. EPI is one of the fastest imaging pulse sequences, allowing image acquisition in the order of a few tens of ms. Thus, it plays an essential role in many advanced MR applications, such as diffusion, perfusion, neuro-functional, and dynamic imaging. However, EPI sequences are prone to artifacts due to T_2^* decay, eddy currents, flow, and off-resonance spins (more details in Chapter 12).

1.4.6.2 Non-Cartesian trajectories

There are advantages in collecting data using non-Cartesian (non-rectangular) k -space trajectories. Radial imaging, or projection reconstruction (PR), was the original MR k -space trajectory used by Paul C. Lauterbur to make the first MR image [11]. To perform the radial projection pattern, frequency encoding is employed together on multiple axes (e.g., x and y for transverse slices) with various amplitudes to attain a rotational pattern of radial spokes (Fig. 1.13c) [6].

Another imaging trajectory is commonly known as spiral imaging (Fig. 1.13d) [1,15]. In spiral imaging, long readout trains allow frequency encoding of a spiral arm. This bears potential advantages. First, as in radial imaging, low spatial frequencies are sampled more densely because the spiral density is highest near the origin of the k -space axis, leading to a higher image signal-to-noise ratio (SNR)

and intrinsic robustness against motion. Second, the gradient *slew rate*, the rate at which the gradient strength has to be altered, is low for spiral scanning. However, errors in the gradient accuracy can lead to deviations from the prescribed non-Cartesian trajectories, with the risk of deteriorating the image quality.

Since the radial and spiral k -space samples do not lie on a regular grid, it is not possible to directly employ the fast Fourier transform, and advanced reconstruction methods are necessary. Further details regarding non-Cartesian trajectories and reconstruction will be discussed in Chapter 4 of this book.

1.4.7 Pulse sequence types

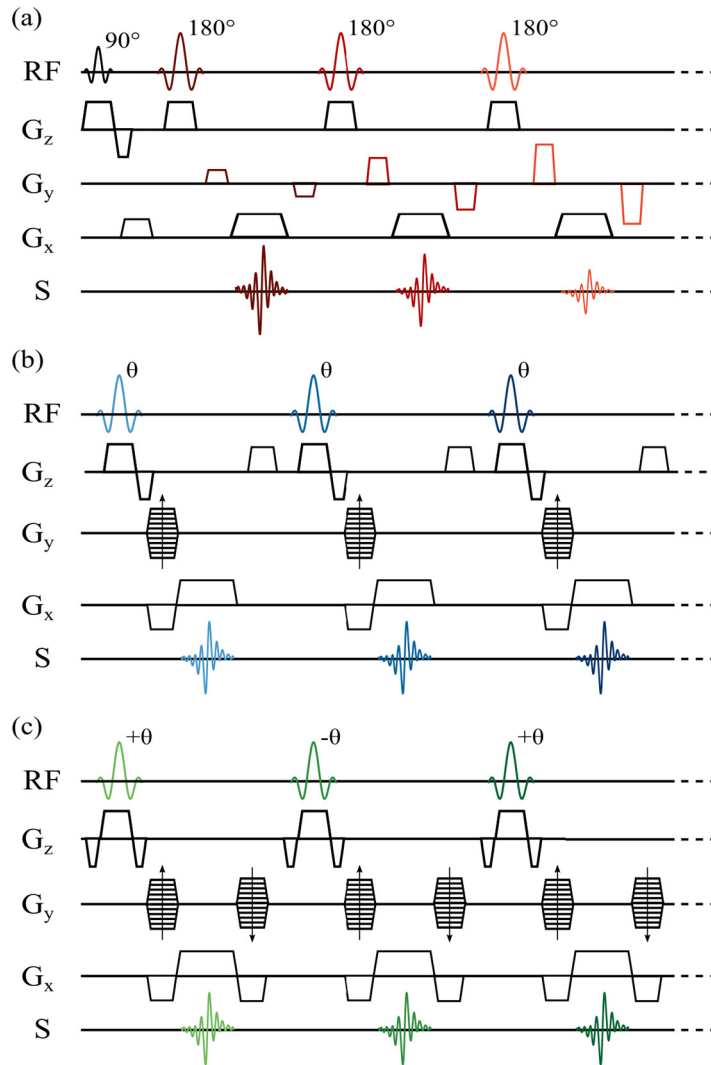
In most cases, numerous excitations are necessary to harvest sufficient information for image reconstruction, as just described. Thus, to allow for efficient imaging, while creating strong magnetization signals with the desired contrast, several pulse-sequence schemes, such as the one shown in Fig. 1.12a, have been devised. In this subsection, we will introduce the most commonly used pulse sequences, divided into two main families: *spin echo* and *gradient echo*. For an in-depth discussion about MRI pulse sequences, the reader is referred to [22].

The difference between *spin echo* and *gradient echo* sequences is most easily explained when considering how a strong signal is being attained: In *spin echo* (SE) sequences the transverse magnetization M_{xy} is being retained, while in *gradient echo* (GRE) sequences, the strong longitudinal magnetization M_z is preserved, as explained now in more detail.

1.4.7.1 Spin echo

In spin echo (SE) sequences, the idea is to keep the transverse magnetization for as long as possible, by refocussing the magnetization dephasing incurred due to the reversible signal decay (T_2' , see Section 1.3.2.2). Thus, for an SE experiment, at least two RF pulses are used for echo generation. First, the transverse magnetization is created with a 90° pulse, which then decays due to dephasing. After time τ , a 180° pulse is played to recover the signal losses incurred due to reversible dephasing, such as those related to inhomogeneities in the main magnetic field. This leads to maximum rephasing at time point 2τ , creating the so-called *spin-echo*. SE sequences are characterized by excellent image quality precisely because the effects of static field inhomogeneities are eliminated. The trade-off is long scan times, which makes the sequence highly sensitive to motion artifacts. For a review of spin-echo basic concepts, the reader is referred to [31].

Fast spin-echo (FSE) sequences have been introduced to considerably shorten the scan times. This is accomplished by delivering several 180° refocusing RF pulses after a single magnetization excitation, as depicted in its sequence diagram in Fig. 1.14a [9]. Multiple phase-encoding gradients with different amplitudes enable the readout of multiple k -space lines for each excitation. This series of spin echoes is called an echo train, and the number of echoes sampled is the echo train length. FSE sequences have a longer TR to deliver as many refocusing pulses as possible before the transverse magnetization is considerably decayed due to T_2 relaxation. FSE sequences find use in a broad spectrum of MRI applications and are consequently a workhorse in today's clinical MRI.

**FIGURE 1.14**

Pulse sequence diagrams of (a) a fast spin echo (FSE), (b) a fast gradient echo, and (c) balanced steady-state free precession (bSSFP) sequence.

1.4.7.2 Gradient echo

In gradient echo (GRE) imaging, the idea is to only use a part of the longitudinal magnetization to generate a transverse magnetization. This leaves substantial longitudinal magnetization, which can then be used to quickly generate a new transverse magnetization.

In GRE, a single RF pulse is used and gradient encoding is applied during the free induction decay. The time point when the gradient prewinder and rewinder amount to a zero moment is often referred to as the *gradient echo* [14,5]. Every RF excitation chips away from the longitudinal magnetization that can be used for the following TRs. Simultaneously, T_1 relaxation during this time leads to the rebuilding of the longitudinal magnetization. This way, a steady-state magnetization (M_{ss}) is being approached after a number of TRs.

Since no rephasing of the transverse magnetization is applied in GRE sequences, the signal intensity decays much faster than in SE sequences, and the TE has to be much shorter to yield sufficient signal intensity.

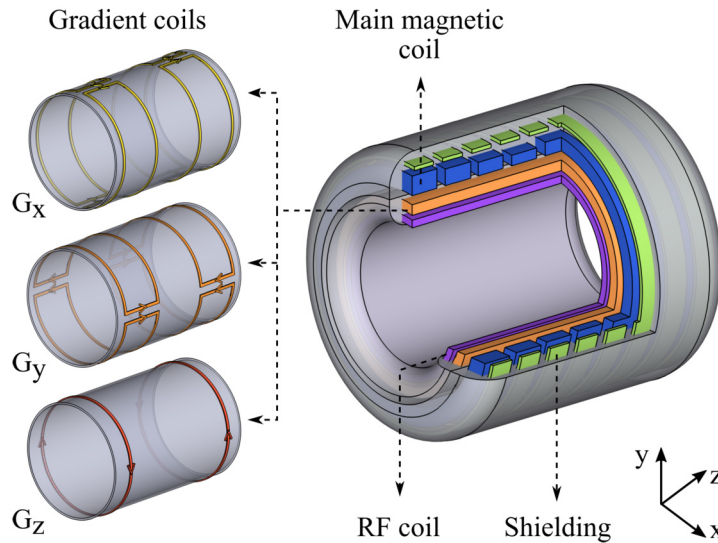
The main advantage of GRE sequences is the ability to achieve short TR and, therefore, faster image acquisition. With very short TRs, however, part of the signal will be “left over” from cycle to cycle. This residual magnetization needs to be accounted for to avoid image artifacts. Two basic concepts are often used for this purpose: 1) destroying (described in the next subsection) or 2) recycling (described in Section 1.4.7.3) the left-over magnetization.

The purposeful destruction of the residual MR signal is called spoiling. Spoiling can be accomplished by turning on additional gradients, often along the slice encoding direction, to dephase the magnetization before the next RF pulse is applied. Alternatively, cycling of the RF phase, i.e., the direction of the B_1 field in the RFR, can be used to achieve spoiling [20]. Both yield a T_1 -weighted image since the transverse magnetization decay does not contribute to the steady-state magnetization. Spoiled GRE sequences are widely used in the clinical setting, but small flip angles, typically much less than 45° , need to be applied for sufficient SNR.

1.4.7.3 *Balanced steady-state free precession*

The idea of recycling the transverse magnetization in a GRE sequence is called a balanced steady-state free precession (bSSFP) sequence [3]. In bSSFP sequences, it is ensured that, at the end of each cycle, the phase incurred by all gradients is fully rewound. This is achieved by a symmetric pulse-sequence diagram as shown in Fig. 1.14c, with identical gradients before and after the gradient echo. This way, a second gradient echo is created at the time point of the RF pulse. Additionally, the RF pulses are played with alternating polarity. In the steady-state, this leads to alternating periods of dephasing and rephasing. As such, in an in-depth analysis, bSSFP sequences can be considered a hybrid of GRE and SE sequences, since both the longitudinal and the transverse magnetization are retained. For more intuition on this, the reader is referred to [23,25].

In bSSFP, the generated contrast is determined by the T_2/T_1 ratio (e.g., blood has a high T_2/T_1 ratio and therefore appears bright in bSSFP images). Due to the recycling of the transverse magnetization, bSSFP images have a higher SNR than spoiled GRE images with comparable TRs. bSSFP sequences are characterized by very short scan times and are thus well-suited for vascular imaging and real-time imaging of moving organs, such as the heart. However, the recycling of the transverse magnetization is prone to differences in the precession frequency of the spins and, thus, off-resonance artifacts are common at high field strengths.

**FIGURE 1.15**

3D cross-sectional view of a conventional clinical MR scanner, with the various major components. The scanner bore comprises a transmitting body coil (RF coil) and gradient coils, surrounded by a superconducting magnet, responsible for the main magnetic field. The shielding procedure is achieved by placing permanent magnets and shield coils. On the left, the three different types of gradient coils are depicted.

1.5 Components of an MRI scanner

Modern MRI systems can be described by three major components, corresponding to three integral parts of the image formation:

1. The superconducting magnet, responsible for the main magnetic field $\mathbf{B}_0$ that facilitates the formation of the net magnetization [26,24];
2. Radiofrequency (RF) coils, placed around and/or on the subject, inducing secondary magnetic fields to create and detect transverse magnetization [27];
3. Three sets of magnetic field gradient coils, enabling spatial encoding of the signal [34,30].

Fig. 1.15 contains a schematic representation of these components.

1.5.1 Magnet

The design of a magnet has to consider several factors to create a magnetic field $\mathbf{B}_0$ with maximal spatial homogeneity and temporal stability in the subject [30]. These factors are important to avoid inhomogeneity-related artifacts and to achieve the best possible SNR. Over the years, magnet design has evolved to enable stronger and more homogeneous magnetic fields. A wide variety of designs is available for a vast range of field strength (0.02–10.5T, for whole-body magnets). The magnets can be divided into three types: resistive, permanent, and superconducting.

Resistive magnets are no longer used for clinical MR scanners because the resulting magnetic field is greatly temperature-dependent and the maximum attainable field strength is very low (about 0.2T). In resistive magnets, the magnetic field is created by the passage of an electric current through an inductive loop (typically, a copper wire or aluminum band) and is proportional to the magnitude of the current. However, due to the electrical resistance in the loop, a large amount of electrical power is dissipated in the form of heat, requiring active water-cooling systems with high power consumption to lower the temperature of the magnet.

Permanent magnets are typically employed in the so-called *open bore* systems, which reduce the subject's stress caused by claustrophobia and have lower costs for the installation and maintenance of the system. These magnets use rare earth alloys (like samarium-cobalt and neodymium-iron-boron) in which large magnetic fields are induced during manufacturing. However, the maximum achievable field strength is lower than 1T. The magnetic field is generally vertically oriented with reduced stray fields. However, the field homogeneity and stability are also temperature-dependent, for example, a 1 °C increase will cause a 1,000 ppm decrease in the magnetic field strength of neodymium-iron.

In clinical scanners, which generally operate at a field strength of 1.5T or 3T, superconducting magnets are employed. These magnets use the properties of superconductive materials (like those made from niobium-titanium). Superconductors have zero electrical resistance below a certain material-dependent critical temperature (e.g., -263.96°C (9.2 K) for niobium-titanium) and an associated critical magnetic field (e.g., 10T for niobium-titanium). Thus, once a current is fed into a loop of superconducting wire, it will circulate indefinitely. In comparison with other common types of magnets, superconducting magnets can achieve a much higher field strength (up to 10.5T whole-body, currently) and feature excellent field uniformity.

The superconducting material, which is designed as a conducting copper matrix with multi-stranded filaments, is surrounded by liquid helium at an extremely low temperature (4.2 K). Heat transfer slowly boils off helium, making regular refills necessary (commonly, every 3–18 months). Modern MRI scanners use increasingly effective cryogenic cooling of the helium chamber with the effect of minimizing the boil-off.

To further improve the magnet homogeneity and to minimize any field distortions, fixed shimming can be performed. Permanent magnets and shim coils, an arrangement of independently wired coils, can be employed either passively, actively, or as a combination of both. A more thorough discussion of MRI magnets can be found in [26].

1.5.2 Gradient coils

As discussed in Section 1.4, to spatially encode the MRI signal in three spatial dimensions, three linear magnetic field gradients are used. In reality, these are generated by using *magnetic field gradient coils* or in short *gradient coils*. The gradient coils are placed inside the bore of the magnet, on a cylindrical former. By design, the center of the gradient coils, and also the center of the magnet, is placed at the *isocenter* ($z = 0$, $y = 0$, $x = 0$).

The design of the gradient coils needs to balance four major objectives: good linearity over the imaging field-of-view (FOV), maximum achievable amplitude, minimal rise time, and minimization of eddy-current effects induced in the magnet.

Contrary to the magnet design, where the goal is to obtain a uniform magnetic field, the magnetic field produced by the gradient coils should be as linear as possible. Although the method of producing the magnetic field is the same—current passing through conducting wires—the chosen material,

the cooling system, and the geometrical configuration differ from the main magnet. For the selected material of the three gradient coils, copper at room temperature produces a sufficiently strong gradient (40–80 mT/m for clinical scanners). To minimize the heat losses in this resistive coil design and to maintain the desired temperature, water cooling is used. Although the gradient coils are placed in a cylindrical form inside the bore of the magnet, the geometrical configuration is different for the longitudinal (z) and the transverse (x and y) fields as seen in Fig. 1.15. The z gradient is typically arranged in a *Maxwell pair*, where a pair of wire loops with counter-rotating currents are separated by $r\sqrt{3}$, with r being the coil radius. The x and y gradients use a different layout known as the *saddle coil* arrangement or the *Golay* configuration. However, the coil configuration in x is rotated by 90° with respect to y . With those design considerations, all three gradient fields are approximately parallel to $\mathbf{B}_0$.

For gradient design, as important as the gradient linearity is the minimization of the gradient slew rate, which relates to the times at which a gradient can be turned on and off. The slew rate is commonly defined as the maximum gradient strength of the gradient divided by the rise time. When the gradients are rapidly turned on and off, eddy currents will be induced in the nearby conducting components of the scanner and may be manifested in the final images as specific artifacts and/or signal loss. To reduce the eddy currents, clinical MRI scanners make use of active shielding, where additional coils are placed in the surroundings of the gradient coils, to minimize the stray gradient fields. This, however, results in reduced bore space and additional power consumption. For an in-depth discussion on gradient coil design, the reader is referred to [24,30,34].

1.5.3 Radiofrequency coils

Radiofrequency (RF) coils are the components responsible for generating the rotating magnetic field $\mathbf{B}_1$ and for picking up the signal induced by the rotating magnetic fields in the sample. The RF coils can be divided into two types: *transmit* and *receive*.

Transmit coils

In most clinical scanners, the principal transmitting coil is known as the body coil, which encompasses the full subject volume and is built into the scanner bore. This coil typically uses a saddle or birdcage design, with the latter generally being able to create a more homogeneous distribution of the transmitted energy throughout the object. The transmit RF coils commonly extend over a large area to optimize the uniformity of the $\mathbf{B}_1$ field and to cover a substantial volume. As a disadvantage, if used as a receiver, such coils will have low sensitivity to the MR signal. When scanning certain anatomies, e.g., the head or the knee, specifically designed coils are used for transmission where less power is required, often at the expense of less uniform fields.

At high field strength, the wavelength of the transmit field approximates the size of the object, which can lead to inadvertent energy distribution, e.g., due to the formation of standing waves. Therefore, in modern MRI systems at high (3T) and, in particular, ultra-high field strengths (7T and above), multiple transmit channels are used simultaneously to circumvent this problem. In a process called B_1^+ shimming, those transmit channels are combined to create optimal homogeneity in the transmit-field amplitude throughout the object and are essential to achieve reliable image quality.

Receive coils

In contrast to transmit coils, dedicated receive coils are designed to better match the anatomy of interest and are placed close to the sample of interest. This will result in increased sensitivity and noise minimization. Generally, two types of receive coils can be differentiated: *volume* and *surface* coils.

Volume coils, which are typically combined transmit and receive (Tx/Rx), cover the full area of interest. In contrast, surface coils are placed on top of the patient's anatomy of interest (e.g., a flexible surface coil is positioned on the chest when imaging the cardiac or abdominal region) to detect signals from a patient's specific anatomical region. Generally, surface coils are only used for signal reception because the RF field response with this coil design is highly inhomogeneous.

To circumvent the limited FOV, an array of multiple surface coils can be combined and used simultaneously to receive the MR signal, while maintaining the high sensitivity of each element. To eliminate the interaction between the different individual elements, these elements need to be decoupled. Most commonly, the coil elements are decoupled by design (i.e., there is a geometrical overlap between neighboring elements). Since each element acts as an independent receiver, the coil noise⁵ is uncorrelated with the other elements in the coil. Images from multiple coil elements are then combined in the reconstruction to yield coverage of the entire imaging volume. Additional details about multiple coil reconstruction will be described in Chapter 6. In-depth analysis of MRI RF coils can be found in [27].

1.5.4 Noise properties

As with any measurement, the quality of the MRI signal and the resulting images are curtailed by noise in the measurement process [29]. Multiple sources of noise taint the signal quality. The overall measurement is usually characterized by the dimensionless signal-to-noise ratio (SNR). To maximize SNR, either the signal strength needs to be improved or the noise contributions need to be mitigated.

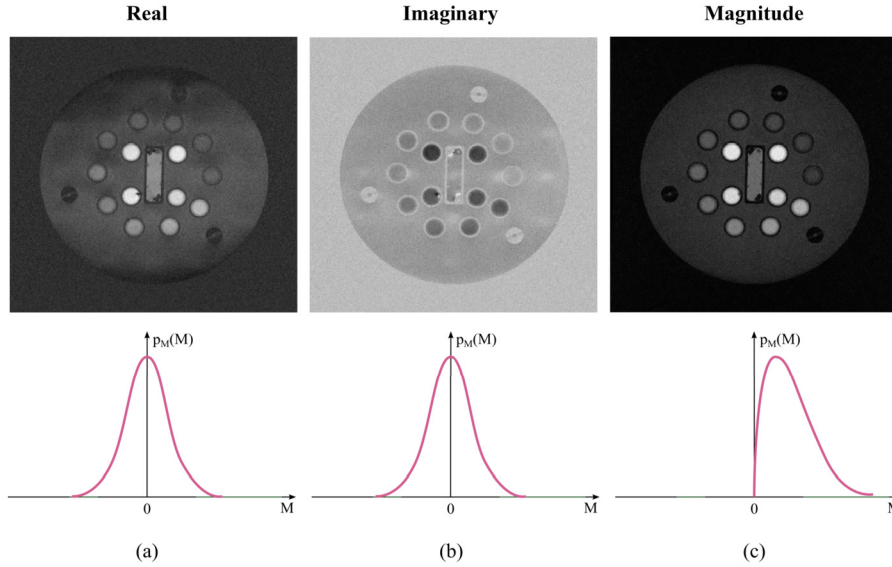
Even though the MR signal is complex, most commonly the magnitude images are used for clinical diagnosis (Fig. 1.16). The mapping from real and imaginary components to their magnitude is nonlinear. Therefore, even though noise in the real and imaginary components can be assumed to follow a Gaussian distribution, noise in the magnitude images is non-Gaussian. For the simplest case of receiving the signal with one channel, and without using advanced reconstruction methods, the noise distribution for the measured pixel intensity M is given by the Rice distribution

$$p_M(M) = \frac{M}{\sigma^2} e^{-\frac{(M^2 + A^2)}{2\sigma^2}} I_0\left(\frac{A \cdot M}{\sigma^2}\right), \quad (1.56)$$

where A is the image pixel intensity without noise, I_0 is the modified zeroth-order Bessel function of the first kind, and σ is the standard deviation of the Gaussian noise in the real and imaginary images, which can be assumed to be identical [7]. This corresponds to the Rice distribution and can be obtained by taking the square root of the sum of squares of two independent and identically distributed Gaussian variables [19].

With Rician noise, the SNR can be defined as the ratio of A over σ . For high SNR (≥ 3), the noise distribution can be well approximated with a simple Gaussian distribution [7]. However, for SNR

⁵ Unfortunately, this is only one of two types of noise, as elaborated in the next section.

**FIGURE 1.16**

Real (a), imaginary (b), and magnitude (c) images of the NIST/ISMRM system phantom, with their corresponding noise distributions.

values ≤ 1 , the noise distribution is substantially skewed. In regions where only noise is presented (i.e., $A = 0$ in Eq. (1.56)), $p_M(M)$ simplifies to a Rayleigh distribution. Note that multicoil reconstructions or nonlinear reconstructions further influence the noise distribution and can lead to different noise properties.

As described in Eq. (1.5), the net magnetization is proportional to the value of the static magnetic field $\mathbf{B}_0$ since stronger $\mathbf{B}_0$ fields lead to an increase in the nuclear polarization and, thus, an increase in signal. Also, at higher field strength, higher voltage is produced in the coil by the precessing magnetization, which further improves the signal strength. However, when $\mathbf{B}_0$ increases, the value of T_1 increases and T_2^* decreases, resulting in a decreased signal strength. Thus, the choice of an optimal field strength remains highly application-dependent.

Besides the signal strength, two important sources of noise need to be considered: first, the human body, which is conducting and, thus, produces thermal noise. This is picked up by the receiving RF coil during imaging. Second, random voltage variations in the copper conductors in the RF coil itself will also increase the received noise. The SNR can be considered proportional to the sensitivity of the RF coil.

Several other imaging parameters affect the SNR. If multiple images are obtained using equal parameters, averaging can be performed to enhance the SNR at the cost of extra scan time. Since the MRI signal is coherent but the noise is incoherent, the SNR increases with $\sqrt{N_a}$, where N_a is the number of signal averages. Also, as the imaging signal is proportional to the number of hydrogen nuclei per voxel, an increase in voxel size proportionally increases the SNR. Accordingly, for a given FOV, if the amount of phase-encoding steps N_{pe} is doubled, then the image SNR is decreased by a factor of two.

Similarly, if the number of points collected in the frequency-encoding direction N_f is doubled for a given FOV, the SNR decreases by a factor of two. If frequency-encoding is performed with twice the bandwidth, the SNR is decreased by a factor of two, leading to shorter readouts but more noise in the signal. Finally, increasing the slice thickness yields a proportional improvement in the image SNR as the imaging volume and the number of protons in the volume are increased.

1.6 Summary

This chapter introduces the basic aspects of MRI physics and the basic concepts involved in image acquisition. This will hopefully enable the reader to understand advanced MRI methods and image-reconstruction techniques as described in the remainder of this book. It describes the physics of MRI around six central principles:

1. *Origin of the net magnetization:* Certain nuclei are characterized by a physical property, a nonzero spin, which causes them to form a magnetic moment when placed into an external magnetic field. Many of these nuclei together form a net magnetization, which is the basis of all MRI signals.
2. *The dynamics of this magnetization in an external magnetic field:* The net magnetization is related to both a magnetic and angular momentum. Thus, if the net magnetization is angled with respect to the external effective field, it prescribes a precession motion around the axis of the effective field, with a characteristic field strength-dependent resonance frequency, the so-called Larmor frequency.
3. *Using rotating magnetic fields to create a detectable magnetization component:* While a strong static magnetic field is necessary to create net magnetization, a much weaker rotating magnetic field can be used to change the orientation of the effective field. Specifically, when this second magnetic field is applied near the resonance frequency, the net magnetization can be tipped away from the direction of the main magnetic field.
4. *Using induction to measure this component:* A magnetization component orthogonal to the main magnetic field precesses and can be used to create a signal. Specifically, this temporally varying magnetic field induces a current in the nearby receiver coils via induction, which can be recorded as the imaging signal.
5. *The evolution of this signal in the presence of relaxation mechanisms:* If the net magnetization is altered from its equilibrium state, e.g., by nutating away from the main magnetic field, multiple recovery processes are at play to return the magnetization back to the thermal equilibrium. The longitudinal magnetization, i.e., the component along the axis of the main magnetic field, undergoes an exponential recovery. The transverse magnetization, which is the component orthogonal to the main magnetic field, experiences an exponential decay.
6. *Gradient fields to enable image formation:* To spatially locate the acquired signal, gradient fields are used to create a spatially dependent phase in the signal. The signal is then acquired in the so-called k -space. To fully resolve the signal in all spatial dimensions, frequency, phase, and slice-encoding gradients can be used.

Based on these basic principles, a wide variety of pulse sequences that allow for acquisitions with different properties can be designed. Most fundamentally, gradient-echo and spin-echo sequences have been introduced along with their variations to make possible faster imaging. Image acquisition can also be performed by sampling various trajectories in k -space to further speed up the scan.

For the practical realization of the imaging process, several hardware components relating to these physical principles are necessary. Superconducting electromagnets are most widely used to create the primary magnetic field. Different coil arrangements enable spatially varying gradient fields and the creation of a secondary magnetic field to excite the magnetization. Anatomy-specific coil arrays are used to receive the signal at maximum strength.

The basic properties of the magnetization signal, its dynamics, and relaxation help to understand the limitations in MRI pulse-sequence design. The fundamental spatial encoding principles and the related hardware further explain the bottlenecks in the image acquisition speed. In recent years, advanced reconstruction methods have been taking center stage in the acceleration of MRI scans. Craftily reducing the number of samples, or supplementing the acquired information with a-priori knowledge has enabled unprecedented acquisition times. However, in MRI, the development of reconstruction techniques, as described throughout the remainder of the book, goes hand in hand with a thorough understanding of the physical principles of the magnetization signal. This chapter is intended to serve as a reference for understanding the limitations and opportunities in the signal properties of MRI that are relevant for image reconstruction.

References

- [1] Ahn CB, Kim JH, Cho ZH. High-speed spiral-scan echo planar nmr imaging-I. *IEEE Trans Med Imaging* 1986;5(1):2–7.
- [2] Bloch F, Hansen WW, Packard M. Nuclear induction. *Phys Rev* 1946;69:127.
- [3] Carr H. Steady-state free precession in nuclear magnetic resonance. *Phys Rev* 1958;112(5):1693.
- [4] Carr HY. Free precession techniques in nuclear magnetic resonance. PhD. 1953.
- [5] Elster AD. Gradient-echo mr imaging: techniques and acronyms. *Radiology* 1993;186(1):1–8.
- [6] Glover GH, Pauly JM. Projection reconstruction techniques for reduction of motion effects in mri. *Magn Reson Med* 1992;28(2):275–89.
- [7] Gudbjartsson H, Patz S. The Rician distribution of noisy MRI data. *Magn Reson Med* 1995;34(6):910–4.
- [8] Hahn EL. Spin echoes. *Phys Rev* 1950;80:580–94.
- [9] Hennig J, Nauerth A, Friedburg H. Rare imaging: a fast imaging method for clinical mr. *Magn Reson Med* 1986;3(6):823–33.
- [10] Larmor J. LXIII. On the theory of the magnetic influence on spectra; and on the radiation from moving ions. *Philos Mag* 1897;44(271):503–12.
- [11] Lauterbur PC. Image formation by induced local interactions: examples employing nuclear magnetic resonance. *Nature* 1973;242(5394):190–1.
- [12] Mansfield P. Multi-planar image formation using NMR spin echoes. *J Phys C, Solid State Phys* 1977;10(3):L55–8.
- [13] Mansfield P, Grannell PK. NMR diffraction in solids. *J Phys C, Solid State Phys* 1973;6(22):L422.
- [14] Markl M, Leupold J. Gradient echo imaging. *J Magn Reson Imaging* 2012;35(6):1274–89.
- [15] Meyer CH, Hu BS, Nishimura DG, Macovski A. Fast spiral coronary artery imaging. *Magn Reson Med* 1992;28(2):202–13.
- [16] Purcell EM, Torrey HC, Pound RV. Resonance absorption by nuclear magnetic moments in a solid. *Phys Rev* 1946;69:37–8.
- [17] Rabi II. Space quantization in a gyrating magnetic field. *Phys Rev* 1937;51:652–4.
- [18] Rabi II, Zacharias JR, Millman S, Kusch P. A new method of measuring nuclear magnetic moment. *Phys Rev* 1938;53:318.

- [19] Rice SO. Mathematical analysis of random noise. *Bell Syst Tech J* 1944;23(3):282–332.
- [20] Zur Y, Wood M, Neuringer L. Spoiling of transverse magnetization in steady-state sequences. *Magn Reson Med* 1991;21(2):251–63.

Suggested readings

- [21] Abragam A. The principles of nuclear magnetism. Oxford University Press; 1961. p. 32.
- [22] Bernstein MA, King KF, Zhou XJ. Handbook of MRI pulse sequences. Elsevier; 2004.
- [23] Bieri O, Scheffler K. Fundamentals of balanced steady state free precession mri. *J Magn Reson Imaging* 2013;38(1):2–11.
- [24] Chapman B. Shielded gradients. And the general solution to the near field problem of electromagnet design. *Magn Reson Mater Phys Biol Med* 1999;9(3):146–51.
- [25] Chavhan GB, Babyn PS, Jankharia BG, Cheng H-LM, Shroff MM. Steady-state mr imaging sequences: physics, classification, and clinical applications. *Radiographics* 2008;28(4):1147–60.
- [26] Cosmus TC, Parizh M. Advances in whole-body MRI magnets. *IEEE Trans Appl Supercond* 2010;21(3):2104–9.
- [27] Gruber B, Froeling M, Leiner T, Klomp DW. RF coils: a practical guide for nonphysicists. *J Magn Reson Imaging* 2018;48(3):590–604.
- [28] Hanson LG. Is quantum mechanics necessary for understanding magnetic resonance?. *Concepts Magn Reson, Part A. Educ J* 2008;32(5):329–40.
- [29] Henkelman RM. Measurement of signal intensities in the presence of noise in MR images. *Med Phys* 1985;12(2):232–3.
- [30] Hidalgo-Tobon S. Theory of gradient coil design methods for magnetic resonance imaging. *Concepts Magn Reson, Part A* 2010;36(4):223–42.
- [31] Jung BA, Weigel M. Spin echo magnetic resonance imaging. *J Magn Reson Imaging* 2013;37(4):805–17.
- [32] Ljunggren S. A simple graphical representation of Fourier-based imaging methods. *J Magn Reson* (1969) 1983;54(2):338–43.
- [33] Rabi II, Ramsey NF, Schwinger J. Use of rotating coordinates in magnetic resonance problems. *Rev Mod Phys* 1954;26(2):167.
- [34] Turner R. Gradient coil design: a review of methods. *Magn Reson Imaging* 1993;11(7):903–20.
- [35] Twieg DB. The k-trajectory formulation of the NMR imaging process with applications in analysis and synthesis of imaging methods. *Med Phys* 1983;10(5):610–21.